MyeliMetric-V2.1 User Manual

Nov 9, 2026 · @Intakhar Ahmad

1. About MyeliMetric-V2.1

MyeliMetric-V2.1 is a Windows application for standardized g-ratio analysis of myelinated fibers from post-segmented axon and myelin measurements. It keeps the complete Version 2 workflow and adds statistics that treat the animal, not the individual fiber, as the experimental unit.

Fibers measured in the same animal share its genotype, treatment, tissue processing and imaging, so they are not independent replicates. Earlier versions compared groups with every fiber counted as a replicate, which could overstate significance. V2.1 corrects this with a nested analysis.

What each version provides

Version	Main features
V1 (published, ASN Neuro 2025)	G-ratio and fiber-diameter calculation, exclusion thresholds, six fiber-diameter bins from the first control animal, per-bin descriptive statistics, Shapiro-Wilk normality testing, figures
V2	Standalone Windows application, three-step workflow, one save location with organized output folders, adjustable bins and thresholds, bin ranges from all pooled control animals, CSV input, run logs, exclusion audit
V2.1	Everything in V2, unchanged, plus: nested split-plot ANOVA with the animal as the unit, animal mixed-effects model with optional slide or section nesting, mean or median bin summaries (median matches GraphPad Prism), bin-free caliber model, animal-level group test, main result figure and a dedicated results workbook

On the V2 example files, every Control, Experimental and Summary workbook produced by V2.1 is identical to the V2 output. The new statistics are written separately to the Summary/Statistics folder and to the main result figure.

Definitions

- **Axon diameter (Ax)**: inner diameter of the axon, in micrometers.
- **Myelin (My)**: the full myelin contribution across the fiber, that is, twice the radial myelin thickness, in micrometers.
- **Fiber diameter** = Ax + My.
- **G-ratio** = Ax / (Ax + My).

2. Installation and starting the program

MyeliMetric-V2.1 runs on Windows 10 or 11 (64-bit). It can be used as a standalone application or run from the Python source code.

Option A: standalone application

1. Download `MyeliMetric-V2.1.exe` from the latest release of the GitHub repository: https://github.com/Intakhar-Ahmad/NeuroMyelin-G-Ratio-Analysis-Toolkit
2. Double-click the file. No installation is needed.
3. If Windows SmartScreen shows a warning, choose More info, then Run anyway.

Option B: run from source

1. Install 64-bit Python 3.11 or 3.12 from python.org.
2. Download and unzip `MyeliMetric-V2.1.zip`.
3. Double-click `RUN_FROM_SOURCE.bat`. The first start creates a local environment and installs the required packages (internet needed once).

Alternatively, from a command prompt in the folder: `pip install -r requirements.txt`, then `python MyeliMetricV2.1.py`.

Option C: build the application yourself

Double-click `BUILD_WINDOWS_EXE.bat`. It creates a build environment, installs the requirements, runs the 28 scientific tests and, only if all pass, builds `dist\MyeliMetric-V2.1.exe`.

Checking an installation

Run `MyeliMetric-V2.1.exe --selftest C:\some\empty\folder` from a command prompt. The program analyzes the bundled example data and writes `SELFTEST_RESULT.txt`, which contains PASS when the installation works.

Package contents

File or folder	Purpose
`MyeliMetricV2.1.py`	Program start file (graphical interface)
`myelimetric/`	Analysis engine: input reading, processing, statistics, figures, reports
`example_Dataset/`	Example data and the long-format template
`tests/`	Scientific test suite
`requirements.txt`	Python packages required
`RUN_FROM_SOURCE.bat`, `BUILD_WINDOWS_EXE.bat`, `MyeliMetric.spec`	Start and build scripts

File or folder	Purpose
`README.md`, `CHANGELOG.md`, `LICENSE`	Documentation and license

3. Preparing input data

MyeliMetric-V2.1 reads Excel (.xlsx, .xls) or CSV files containing one measured axon diameter and one myelin value per fiber, in micrometers. Two layouts are accepted; the program recognizes the layout automatically.

Wide layout (published format)

One file per group. Each animal has a pair of columns with the same identifier before the underscore: `<ID>_Ax` and `<ID>_My`.

```
CTL1_Ax,CTL1_My,CTL2_Ax,CTL2_My,CTL3_Ax,CTL3_My
0.82,0.28,0.77,0.31,0.69,0.26
0.65,0.24,0.91,0.33,0.74,0.29
```

The experimental file uses the same pattern (`EXP1_Ax, EXP1_My, ...`). Empty spacer columns between animals are ignored, and animals may have different numbers of fibers.

Optional slide or section IDs

To record which slide or section each fiber was measured on, add a column directly after each animal's `_Ax` and `_My` pair. Accepted headers are `Slide-ID`, `Slide`, `Section`, `Section-ID`, `Field` or `Image`, or a named form such as `CTL1_Section`.

```
CTL1_Ax,CTL1_My,Slide-ID,CTL2_Ax,CTL2_My,Slide-ID
0.82,0.28,S1,0.77,0.31,S1
0.65,0.24,S2,0.91,0.33,S2
```

Section IDs are optional. When they are given, they must be given for every animal in both groups, and every measured fiber needs one; otherwise the program stops and names the animals concerned. Section labels only need to be unique within an animal (S1 of CTL1 and S1 of CTL2 are different sections).

Long layout

One file containing both groups, one row per fiber, with the columns `Group`, `Animal`, `Ax`, `My` and, optionally, `Section`. Load it in Step 1 and leave Step 2 empty. A template is provided as `example_Dataset/Template_long_format.csv`.

```
Group,Animal,Section,Ax,My
CTL,M1,S1,0.82,0.28
CTL,M1,S2,0.71,0.26
EXP,M7,S1,0.77,0.19
```

The group named CTL (if present) is used as the reference group; otherwise the first group in alphabetical order is.

Data checks applied on reading

- A column pair with only `_Ax` or only `_My` stops the analysis with a message naming the identifier.
- Rows where both values are empty are treated as padding and ignored.

- Rows with only one of the two values are excluded and listed in the exclusion audit.
- Non-numeric entries are treated as missing.

Example data supplied

File	Content
`Example_CTL_3Animals_With_SlideIDs.xlsx`, `Example_EXP_3Animals_With_SlideIDs.xlsx`	Previously published dataset: 3 control and 3 experimental animals, 120 fibers each on 3 slides
`Example_CTL.xlsx`, `Example_EXP.xlsx`	The same measurements without slide IDs
`Example_long_with_sections_SYNTHETIC.csv`	Simulated long-format data (4 animals per group, 3 sections each) for demonstrating the section model; not real data
`Template_long_format.csv`	Empty-style template for the long layout

4. Analysis settings

The Analysis settings panel at the top of the window holds three values. The defaults follow the published method; Reset to defaults restores them.

Setting	Default	Allowed	Effect
Min axon diameter	0.15 µm	0 or more	Fibers with a smaller axon diameter are excluded
Min myelin thickness	0.03 µm	0 or more	Fibers with a smaller myelin value are excluded
Number of bins	6	1 to 30	Number of fiber-diameter bins (subsets)

Excluded fibers are never deleted silently: each one is listed, with the reason, in `Summary/Exclusion_Audit.xlsx`.

How the bins are defined

The fiber-diameter bins are defined from the control group only, so that experimental fibers are compared with control fibers of the same size.

1. Fiber diameters of all included control fibers (all control animals pooled) are sorted.
2. The bin edges are the quantiles of this pooled distribution that divide it into the chosen number of bins of about equal size, rounded to two decimals.
3. Each bin contains fibers larger than the previous edge and up to and including its own edge, so the bins are contiguous and tied values always fall in the same bin.
4. The same edges are applied to every control and experimental animal.

Experimental fibers smaller than the smallest or larger than the largest control fiber fall outside all bins. They are not placed in a subset and are listed in the exclusion audit.

As in V1 and V2, fiber diameter and g-ratio are rounded to two decimals before binning and statistics. This keeps V2.1 results identical to V2 for the same data.

Why these exclusion cutoffs

The two cutoffs remove measurements that cannot come from a myelinated fiber, while keeping every biologically plausible one. They are the values of the published method.

- **Axon diameter below 0.15 µm.** In the central nervous system, myelination is closely tied to axon caliber: axons are myelinated when they exceed a threshold diameter of about 0.2 to 0.4 µm (Chang, Redmond and Chan, 2016). Axon profiles below 0.15 µm therefore lie under the smallest myelinated axons and usually reflect segmentation errors, grazing sections through a fiber or debris. The cutoff is placed below 0.2 µm on purpose, so that genuine small myelinated axons are retained.
- **Myelin value below 0.03 µm.** The myelin value is the total myelin contribution across the fiber, so 0.03 µm corresponds to 15 nm of myelin on each side of the axon. Compact myelin repeats with a periodicity of about 12 nm per lamella (Simons and Nave, 2016), so a smaller value represents about one lamella or less. Such values are not measurable compact myelin; they usually come from tracing errors or unmyelinated profiles, and would produce spurious g-ratios close to 1.

Both cutoffs are deliberately permissive. In the example dataset no fiber fell below either value (smallest axon diameter 0.20 µm, smallest myelin value 0.05 µm), so they changed nothing. For other tissues, species or imaging resolutions the cutoffs can be changed; always report the values used.

Why six bins

G-ratio depends on fiber caliber, so groups are compared within fiber-size bins: like is compared with like. Six bins is the default of the published method and is kept so that results remain comparable with published analyses. It is also an appropriate size for typical datasets:

- **Enough fibers per animal and bin.** Because the bins hold equal numbers of control fibers, six bins give each animal about one sixth of its fibers per bin. With 120 fibers per animal this is about 20 fibers per animal and bin (10 to 37 in the example dataset). The standard error of a bin mean based on 20 fibers is about 0.22 times the fiber standard deviation, which gives stable animal-by-bin values.
- **Enough size resolution.** Six bins span small to large fibers finely enough to detect a group difference confined to part of the size range.
- **Trade-offs.** More bins mean fewer fibers per animal and bin, less precise values, a higher risk that an animal has no fibers in some bin (which prevents the split-plot ANOVA), and a larger sphericity correction. Fewer bins give coarser size resolution and can hide a size-dependent group effect.

As a guide, choose the number of bins so that each animal has roughly 15 to 20 fibers per bin, that is, about the number of fibers per animal divided by 20:

Fibers per animal	Suggested number of bins
60	3
100	5

Fibers per animal	Suggested number of bins
120	6 (default)
200 or more	8 to 10

This guide is a practical rule of thumb, not a formal requirement. Report the number of bins used. The caliber mixed model (section 8.3) does not use bins and can confirm that a conclusion does not depend on the bin number.

5. Running an analysis

A complete analysis takes five steps. The figure at the end of this section shows the main window.

1. **Choose save location.** Select one folder. All results are written there, in Control, Experimental and Summary subfolders.
2. **Step 1: Load control data.** Select the control file (wide layout) or the single long-layout file containing both groups.
3. **Step 2: Load experimental data.** Select the experimental file. Skip this step for a control-only run or when a long-layout file was loaded in Step 1.
4. **Confirm independent animals.** Tick the box stating that each sample ID (CTL1, CTL2, EXP1 ...) is a different animal.
5. **Step 3: Analyze.**

The analysis runs in the background. Progress and every saved file are reported in the Activity log, and a message appears when the run is complete, with an option to open the output folder.

The independent-animal confirmation

The statistics are valid only if each sample ID is a separate animal. Images, fields or sections from one animal must not be entered as separate sample IDs; they belong in a Slide-ID column (wide layout) or Section column (long layout).

If the box is not ticked when Step 3 is clicked, the program asks directly:

- **Yes**: the box is ticked and the full statistics are run.
- **No**: descriptive results only are saved, without p-values, and `Summary/Statistics/INFERENCE_NOT_RUN.txt` explains why.
- **Cancel**: returns to the window.

Saving into a folder used before

Running again into the same save location replaces the previous MyeliMetric results, so old and new files cannot mix. A folder that already contains Control, Experimental or Summary folders not created by MyeliMetric is refused, to protect other files.

Control-only runs

With control data only, MyeliMetric produces all Control outputs and a summary of the control animals. Group comparisons and statistics need both groups and are skipped.

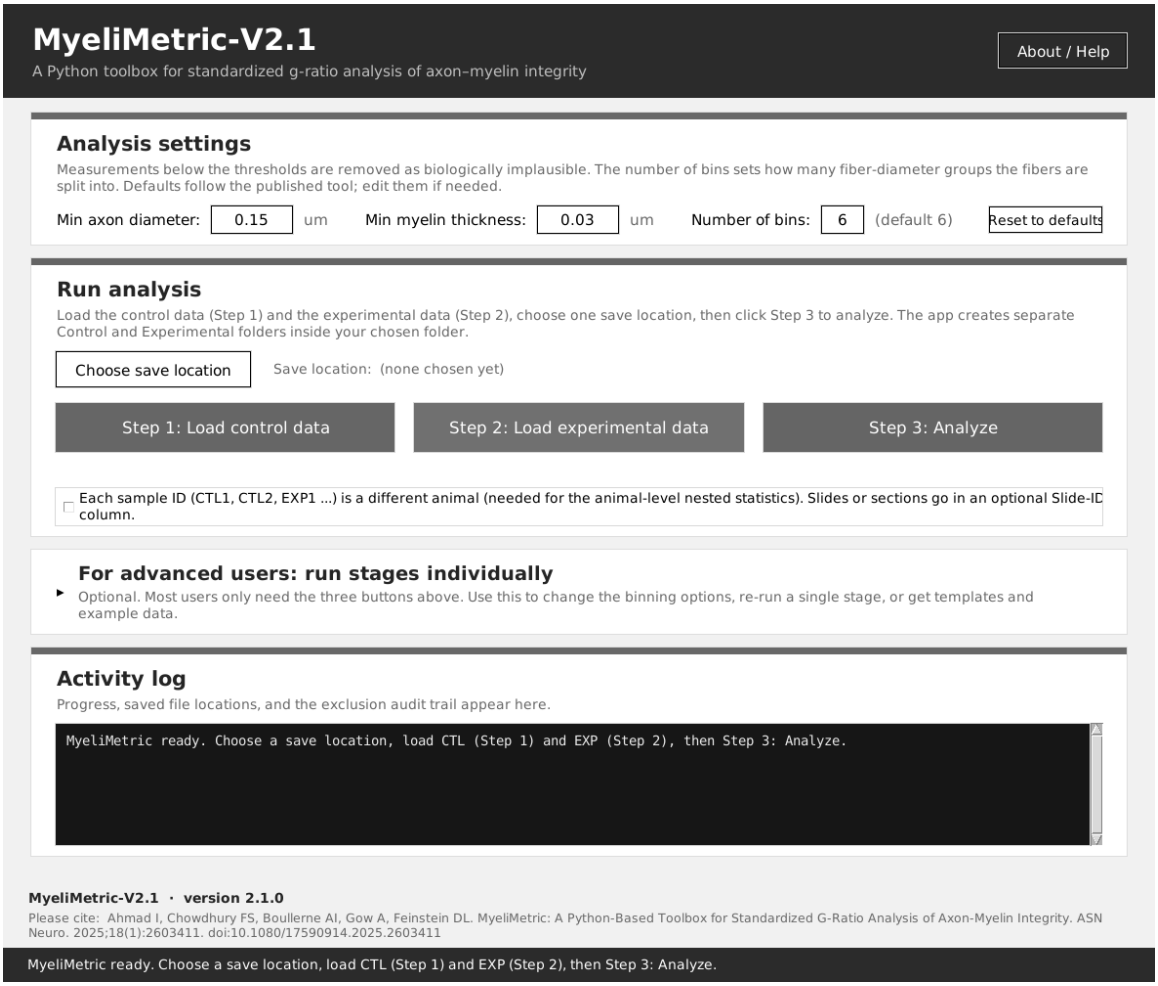


Figure 1. The MyeliMetric-V2.1 main window. From top: Analysis settings; Run analysis with the save location, Steps 1 to 3 and the independent-animal confirmation; the collapsed advanced panel; the Activity log.

6. Advanced options

Click "For advanced users: run stages individually" to open the advanced panel. Most analyses need none of these options.

Analysis options

Option	Choices	Default	When to change it
Bin fibers by	Fiber diameter (published); Axon diameter	Fiber diameter	Axon diameter avoids binning on a quantity that contains the myelin being compared (see section 8)
Bin ranges from	Pooled controls (V2 default); First control animal	Pooled controls	First control animal reproduces the bin ranges of the published V1 analysis

Option	Choices	Default	When to change it
	(published V1)		
Nested ANOVA uses each animal's bin	Mean (default); Median (as in Prism)	Mean	Median matches the GraphPad Prism mixed-effects workflow and is less sensitive to outlying fibers
Save histograms	On; Off	On	Turn off to speed up large runs

Mean or median

For the nested ANOVA, all fibers of one animal in one bin are summarized by a single value. With Mean, this is their mean g-ratio; with Median, their median g-ratio. The median option changes:

- the split-plot ANOVA, which then uses animal-by-bin medians;
- `Final_Result.png`, which then shows medians with interquartile ranges;
- `Nested_ANOVA_Results.xlsx`, which then contains an Animal_x_Bin_Medians sheet;
- the animal-level group test, which then compares the mean of each animal's bin medians.

The V2 descriptive workbooks and figures and the mixed-effects models are the same with either choice.

Stage tools

Button	Result
Clean data only (+ exclusion audit)	Applies the thresholds and writes cleaned workbooks and the exclusion audit to `Cleaned_only/`
Control bin ranges only	Computes the control bin ranges and writes them to `Ranges_only/`
Save long-format template	Saves an example long-layout CSV file
Copy example data	Copies the bundled example files to a folder of your choice
Open output folder	Opens the chosen save location

7. Output files

All results are written to the chosen save location. The main result is `Summary/Final_Result.png`; the statistical results are in `Summary/Statistics/Nested_ANOVA_Results.xlsx`.

```
<save location>/
    Control/
        Cleaned/         Cleaned_CTL.xlsx
        Subsets/         Ranges_CTL.xlsx, Subsets_CTL.xlsx
```

```
        Stats/           Stats_CTL.xlsx
        Histograms/      one PNG per animal and bin
        ScatterPlots/    one PNG per animal
        Figures/         Normality_Overview_CTL.png
        run_log_CTL.txt
    Experimental/        same files for EXP, using the control bin ranges
    Summary/
        Final_Result.png
        Summary_Report.xlsx
        Summary_CTL_vs_EXP.png
        Exclusion_Audit.xlsx
        Statistics/
            Nested_ANOVA_Results.xlsx
            Statistics_Report.xlsx
            Results_Summary.txt
            Grand_Mean_GRatio.png
            Animal_Level_Plot.png
            Caliber_Model_Plot.png
    MyeliMetric_run_log.txt
    settings.json
```

Per-group outputs (as in V2)

File	Content
`Cleaned_<group>.xlsx`	Fiber diameter, Ax, My and g-ratio for every animal; excluded fibers left blank
`Ranges_CTL.xlsx`	Minimum and maximum fiber diameter of each control bin
`Subsets_<group>.xlsx`	One sheet per animal and bin with the fibers in that bin
`Stats_<group>.xlsx`	Per animal and bin: median, mean, SD, 95% confidence interval, Shapiro-Wilk statistic, p-value and normality call
Scatter plots	Fiber diameter against axon diameter per animal, with the regression line
Histograms	G-ratio distribution per animal and bin
`Normality_Overview_<group>.png`	Shapiro-Wilk p-value for every animal and bin, with the 0.05 threshold
`run_log_<group>.txt`	Every step of the group's analysis, with counts of included and excluded fibers

Summary outputs

File	Content

File	Content
`Final_Result.png`	Main result figure: each animal's mean (or median) g-ratio in each bin, with group grand means (section 9)
`Summary_Report.xlsx`	Per_Subset: all per-bin statistics of both groups. Grand_Means: grand mean g-ratio per group (as in V2)
`Summary_CTL_vs_EXP.png`	The V2 summary figure, unchanged
`Exclusion_Audit.xlsx`	Counts of read, included and binned fibers per animal and reason; every excluded measurement with its source row

Statistics outputs

File	Content
`Nested_ANOVA_Results.xlsx`	About: design and settings. Nested_ANOVA: Group, Bin and Group x Bin tests. Mixed_Model: animal mixed-effects result and variance shares. Animal_x_Bin_Means (or Medians): the values the ANOVA tested, one row per animal
`Statistics_Report.xlsx`	All analyses with exact values, including the animal-level test and caliber model; a README sheet and a Diagnostics sheet that explains any analysis that could not be run
`Results_Summary.txt`	Draft methods paragraph and results sentences for a manuscript. Edit before use
`Grand_Mean_GRatio.png`	Grand mean per group with each animal shown
`Animal_Level_Plot.png`	One point per animal, as used in the group test
`Caliber_Model_Plot.png`	G-ratio against axon diameter with the fitted group lines

Run records

`MyeliMetric_run_log.txt` records the whole run, and `settings.json` stores the program version, the settings and the input file names, so that every analysis can be reproduced.

8. Statistical methods

All inferential statistics in V2.1 use the animal as the experimental unit. Fibers, and sections when given, are technical subsamples nested within animals; animals are nested within groups. The number of animals is the biological n.

```
Group  >  Animal  >  Section (optional)  >  Fiber
```

8.1 Nested split-plot ANOVA (main analysis)

The fibers of each animal in each bin are summarized by one value, their mean (default) or median. These animal-by-bin values are analyzed with a split-plot (mixed-design) ANOVA:

Effect	Question	Tested against
Group	Do the groups differ in g-ratio?	Variation between animals within groups; df = 1 and N animals minus 2
Bin	Does g-ratio change with fiber size?	Variation within animals
Group x Bin	Does the group difference depend on fiber size?	Variation within animals

Because the bins are repeated measurements on the same animal, sphericity is not assumed. Greenhouse-Geisser and Huynh-Feldt corrected p-values are reported for Bin and Group x Bin; the Greenhouse-Geisser value is recommended for reporting. Effect sizes are given as partial and generalized eta squared.

Every animal must have at least one fiber in every bin. If a bin is empty for any animal, the ANOVA is not run and the Diagnostics sheet names the animal and bin; use fewer bins, or rely on the mixed-effects models, which do not need complete bins.

8.2 Animal mixed-effects model

To account for the correlation among fibers measured in the same animal, per-fiber g-ratios are also fitted with a linear mixed-effects model: group as a fixed effect and a random intercept for each animal. When slide or section IDs are provided, a random intercept for section nested within animal is added. Section IDs are optional; without them the model uses animal IDs alone, and the animal remains the unit of inference.

The model is fitted by restricted maximum likelihood (REML). The group difference is tested with a t statistic on N animals minus 2 degrees of freedom. The share of the total variance at the animal, section and fiber level is also reported.

8.3 Caliber mixed model

G-ratio normally depends on fiber size. Binning by fiber diameter includes the myelin itself, so a fiber with thinner myelin moves to a smaller bin. The caliber model avoids bins: per-fiber g-ratio is modeled on group, axon diameter and their interaction, with a random intercept and slope for each animal (and a section variance component when sections are given). It reports the group difference at the mean axon diameter, the slope in the reference group, and the difference in slopes, each with a 95% confidence interval and a t test on N animals minus 2 degrees of freedom.

8.4 Animal-level group test

Each animal contributes one value: its mean g-ratio (sections weighted equally), or with the median option the mean of its bin medians. The groups are compared with a two-tailed Student t test

(equivalent to a one-way ANOVA), a Welch t test and an exact permutation test, and the result includes the mean difference with its 95% confidence interval and Hedges' g.

8.5 Small numbers of animals

- Large-sample (z) tests of mixed models give too many false positives with few animals. In simulations with 3 animals per group and no true effect, the N minus 2 degrees of freedom used here gave 3% false positives, against 11% for the z test.
- With 3 animals per group, an exact permutation test cannot give p below 0.1; the report states this whenever it applies.
- The split-plot ANOVA needs at least 2 animals per group; the mixed models need at least 3.

8.6 Descriptive statistics (unchanged from V2)

For every animal and bin with at least 3 fibers: median, mean, standard deviation, 95% confidence interval of the mean and the Shapiro-Wilk normality test. The grand mean of each group is the mean of its animal-by-bin means.

9. Interpreting results

This section works through the bundled example files `Example_CTL_3Animals_With_SlideIDs.xlsx` and `Example_EXP_3Animals_With_SlideIDs.xlsx`, a previously published dataset with 3 control and 3 experimental animals (360 fibers per group, 3 slides per animal), analyzed with the default settings.

The main result figure

`Final_Result.png` (Figure 2) shows each animal separately in each fiber-diameter bin. The horizontal axis gives the upper edge of each control bin in micrometers; each marker is one animal's mean g-ratio in that bin, with the standard deviation of its fibers as the error bar (median and interquartile range with the median option). Dashed lines mark the grand mean of each group. Circles are control animals and squares experimental animals.

Read the figure in three ways:

- **Vertical gap between the groups**: the overall group difference (Group effect).
- **Trend from left to right**: whether g-ratio depends on fiber size (Bin effect).
- **Change in the gap from left to right**: whether the group difference depends on fiber size (Group x Bin interaction).

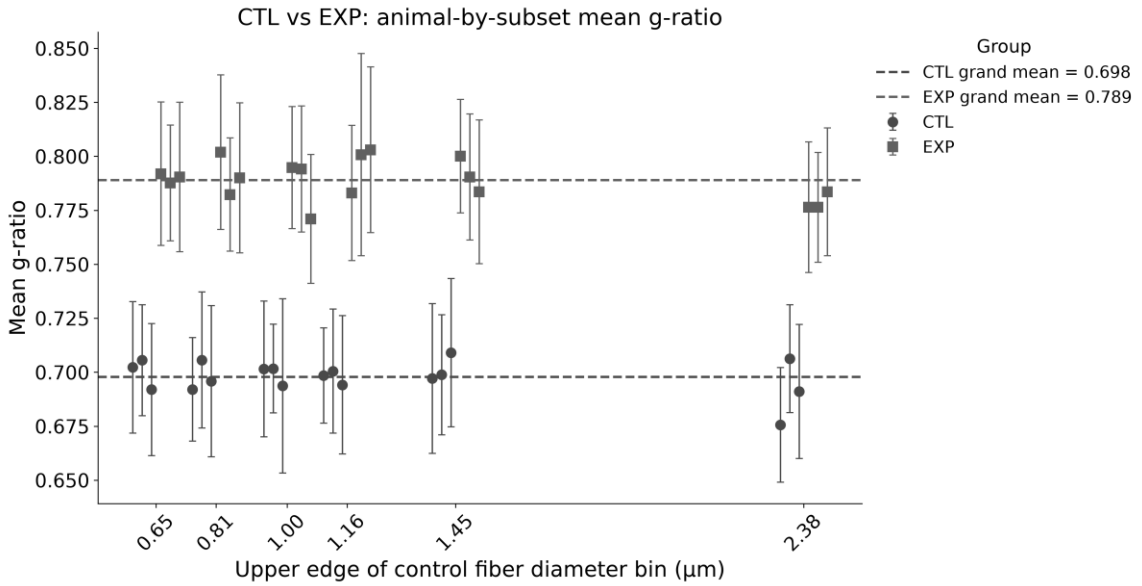


Figure 2. `Final_Result.png` for the example dataset (default settings). Each marker is one animal's mean g-ratio in one fiber-diameter bin (circles: CTL; squares: EXP), with the standard deviation of its fibers. Dashed lines: group grand means (CTL 0.698, EXP 0.789).

Results for the example data

Analysis	Mean (default)	Median option
Grand mean g-ratio, CTL / EXP	0.698 / 0.789	0.696 / 0.790
Group	F(1, 4) = 954.3, p < 0.001	F(1, 4) = 672.0, p < 0.001
Bin (Greenhouse-Geisser)	F(5, 20) = 1.57, p = 0.252	F(5, 20) = 1.05, p = 0.407
Group x Bin (Greenhouse-Geisser)	F(5, 20) = 0.31, p = 0.804	F(5, 20) = 0.86, p = 0.488
Animal mixed model, animal + slide	+0.091 (95% CI 0.083 to 0.100), t(4) = 30.5, p < 0.001	same

Interpretation: experimental animals have a higher g-ratio than control animals (thinner myelin relative to axon size), by about 0.09. The difference is similar across fiber sizes, and g-ratio does not change significantly with fiber size in either group. In the mixed model, slides explained essentially none of the variance beyond the animal (animal 0.6%, slide 0%, fiber 99.4%), so results with and without slide IDs are the same.

Reporting the results

`Results_Summary.txt` contains a draft methods paragraph and results sentences. A typical results statement is:

> *G-ratio was higher in EXP than in CTL animals (grand mean 0.789 versus 0.698; split-plot ANOVA, Group F(1, 4) = 954.3, p < 0.001; n = 3 animals per group). There was no effect*

> *of fiber diameter (Greenhouse-Geisser corrected p = 0.252) and no Group x fiber diameter interaction (p = 0.804).*

Always report the number of animals per group, not only the number of fibers, and state whether bin means or medians were used.

10. Validation

MyeliMetric-V2.1 reproduces V2 exactly and agrees with GraphPad Prism and with independent statistical software. The checks below are part of the 28-test suite that runs before every build.

Check	Reference	Outcome
V2 compatibility	V2 output for the bundled example files	All Control, Experimental and Summary workbooks identical: bin ranges, bin sizes (62, 59, 62, 57, 61, 59), per-bin statistics, Shapiro-Wilk results, grand means 0.6978 and 0.7889
Published V1 bin ranges	First-control-animal option	Bin minima 0.29, 0.68, 0.86, 1.01, 1.17, 1.41 µm, as published
GraphPad Prism	Prism 10.4 two-way repeated-measures mixed-effects analysis (REML) on per-mouse bin medians	Treatment F(1, 4) = 0.3674, p = 0.5772; fiber diameter F = 0.9564, Greenhouse-Geisser epsilon 0.4441, p = 0.4177; interaction F(4, 16) = 0.7118, p = 0.5958; all identical
Prism one-way repeated measures	Prism mixed-effects analysis within control animals	F = 0.2645, epsilon 0.3923, p = 0.7372; identical
Independent ANOVA software	pingouin mixed_anova	F, degrees of freedom, p-values, epsilon and effect sizes identical to 5 decimals
Animal-level test	SciPy t test and one-way ANOVA	Identical
No pseudoreplication	Every fiber duplicated	p-values unchanged
Small-sample error rate	200 simulated datasets, 3 animals per group, no true effect	3% false positives (z test: 11%)
Standalone application	`--selftest` on the bundled data	PASS

Two differences from Prism remain. Prism applies the Greenhouse-Geisser correction to the fiber-diameter factor only; V2.1 also reports a corrected p for the interaction, and its uncorrected p equals Prism's. Prism's mixed-effects fit also accepts empty animal-by-bin cells, whereas the split-plot ANOVA needs every bin filled; with complete data, as in the example, the results are identical.

11. Troubleshooting and frequently asked questions

Message or problem	Cause	Solution
"No valid column pairs were found"	Headers do not follow `<ID>_Ax` and `<ID>_My`	Rename the headers, for example CTL1_Ax and CTL1_My
"These identifiers have only one of _Ax / _My"	A typing difference between the two headers of one animal	Make both identifiers identical
"Section/slide IDs were found for some animals but not for ..."	Slide-ID columns are missing for some animals	Add a Slide-ID column for every animal, or remove all of them
"Section/slide IDs are present in one group's file but not the other"	Only one group has slide IDs	Provide slide IDs for both groups, or for neither
"... already exists and was not created by MyeliMetric"	The save location holds other Control, Experimental or Summary folders	Choose an empty folder or a previous MyeliMetric output folder
No p-values; INFERENCE_NOT_RUN.txt present	The independent-animal confirmation was not given	Tick the box, or answer Yes when asked, and run again
"Every animal needs at least one fiber in every bin"	An animal has no fibers in some bin	Use fewer bins; the mixed models are still reported
"Too many bins for these control values"	Two bin edges coincide	Reduce the number of bins
Experimental fibers missing from subsets	They are smaller or larger than any control fiber	Listed in Exclusion_Audit.xlsx; expected behavior
The window does not open	A required package is missing (source version)	Run RUN_FROM_SOURCE.bat, or `pip install -r requirements.txt`

Should I use means or medians? Means are the default and match V2's descriptive outputs. Medians match the Prism workflow and are less affected by a few unusual fibers. Choose one before analyzing and report which was used.

Do I need slide or section IDs? No. Without them, the analysis uses animal IDs alone and remains valid. With them, the mixed model also estimates how much variation comes from sections.

How many animals do I need? At least 2 per group for the ANOVA and 3 for the mixed models. With 3 animals per group, only large differences can be detected reliably.

Can I enter several images of one animal as separate samples? No. That counts images as animals and inflates significance. Give them one animal ID and record the image or section in a Slide-ID column.

12. Citation, license and contact

If you use MyeliMetric in published work, please cite:

> *Ahmad I, Chowdhury FS, Boullerne AI, Gow A, Feinstein DL. MyeliMetric: A Python-Based Toolbox for Standardized G-Ratio Analysis of Axon-Myelin Integrity. ASN Neuro. 2025;18(1):2603411. doi:10.1080/17590914.2025.2603411*

Please also state the version used. The version is recorded in `settings.json` and in the run log of every analysis.

License. MyeliMetric is open-source software released under the MIT License (see `LICENSE`).

Source code, releases and questions. https://github.com/Intakhar-Ahmad/NeuroMyelin-G-Ratio-Analysis-Toolkit. Please report problems through the repository's Issues page, with the run log attached.

Website. https://intakhar-ahmad.github.io/MyeliMetric-Website/